

A 2-categorical approach to logical relations

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Things I learned from Marcelo

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1. Look for the right
abstractions.

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1. Look for the right **abstractions.**
2. Let the algebra tell you **how things should be.**

“correct-by-construction”

“principled”

“synthesise”

Logical relations,
at varying levels
of abstraction

Level 1: logical relations as families of relations Milner, Plotkin,

A **logical relation** R consists of:

a predicate $R_A \subseteq \llbracket A \rrbracket$ for each type A ...

...determined inductively at higher types by
a **logical relations condition**

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E.g. logical relations condition for $\rightarrow$ in STLC:

$$f \in R_{A \rightarrow B} \subseteq \llbracket A \rrbracket \Rightarrow \llbracket B \rrbracket \quad \text{iff} \quad f(x) \in R_B \text{ whenever } x \in R_A$$

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Basic Lemma:

every program is in the relation: $\llbracket P \rrbracket \in R_A$ for any closed $P : A$

so long as this holds for the basic constants

Level 2: logical relations as relations models

Statman,
Reynolds & Ma,

Level 2: logical relations as relations models

Statman,
Reynolds & Ma,

semantic model

Set

Level 2: logical relations as relations models

Statman,
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objects: $(X \in \mathbf{Set}, R \subseteq X)$

maps: functions preserving the predicate

Pred



semantic model

Set

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objects: $(X \in \mathbf{Set}, R \subseteq X)$

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semantic model

Pred



Set

Pred is cartesian closed:

$$(X, R) \Rightarrow (Y, S) := (X \Rightarrow Y, R \supset S)$$

$$f \in R \supset S \text{ iff } f(x) \in S \text{ whenever } x \in R$$

$$f \in \llbracket A \rightarrow B \rrbracket^{\mathbf{Pred}} \text{ iff } f \text{ preserves the predicate on } \llbracket A \rrbracket^{\mathbf{Set}}$$

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CC-structure encodes the
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Basic Lemma holds automatically:

$$p(\llbracket P \rrbracket^{\mathbf{Pred}}) = \llbracket P \rrbracket^{\mathbf{Set}}$$

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Level 3: logical relations as certain fibrations

Hermida, Jacobs, Katsumata, ...

Level 3: logical relations as certain fibrations

Hermida, Jacobs, Katsumata,

semantic model

$\mathcal{M}$

Level 3: logical relations as certain fibrations

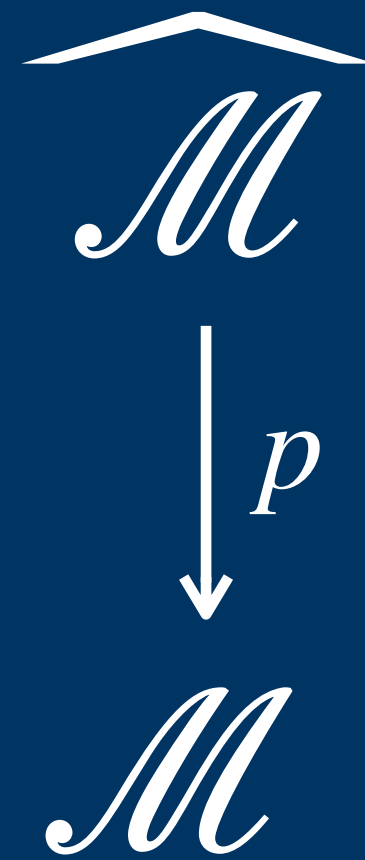
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‘relations model’

forgetful functor p

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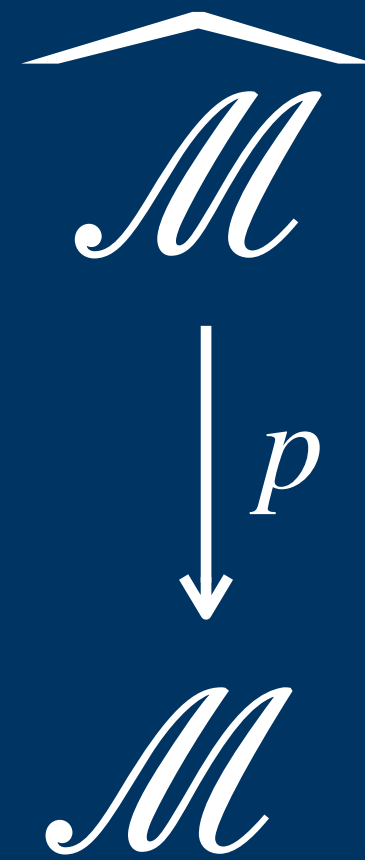
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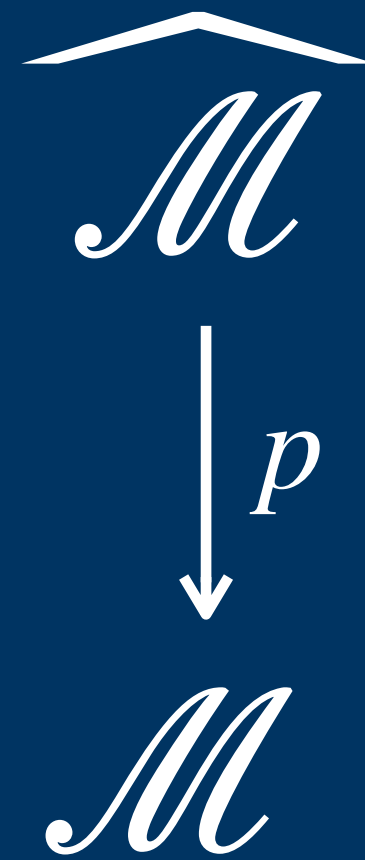
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model-structure encodes the
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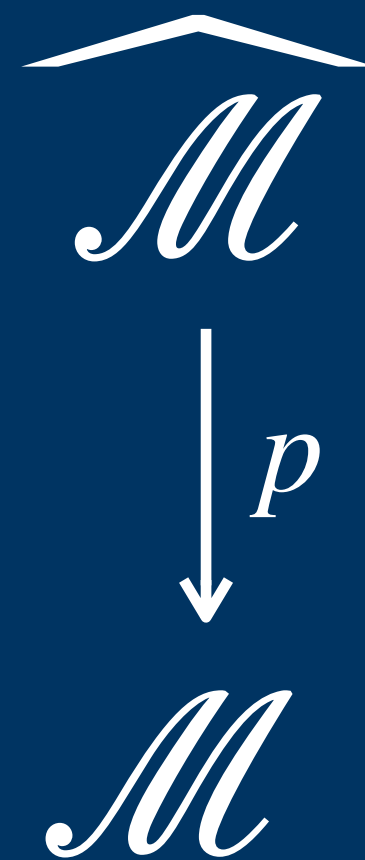
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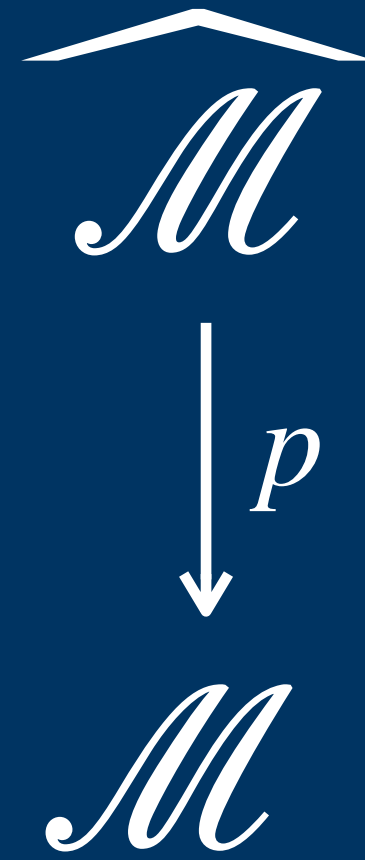
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fibration for logical relations:

fibration
strictly preserving
model structure

Logical relations can be seen as...

Level 1: type-indexed families of relations

Level 2: relations models

Level 3: fibrations for logical relations

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also get:
a good theory for building
new logical relations

fibration for logical relations:
fibration
strictly preserving
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Building fibrations for logical relations

e.g. effect simulation, Π -lifting, Kripke relations of varying arity, ...

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$\mathcal{M}$

$\mathcal{E}$

$\downarrow p$

$\mathcal{B}$

abstract notion
of ‘relation’

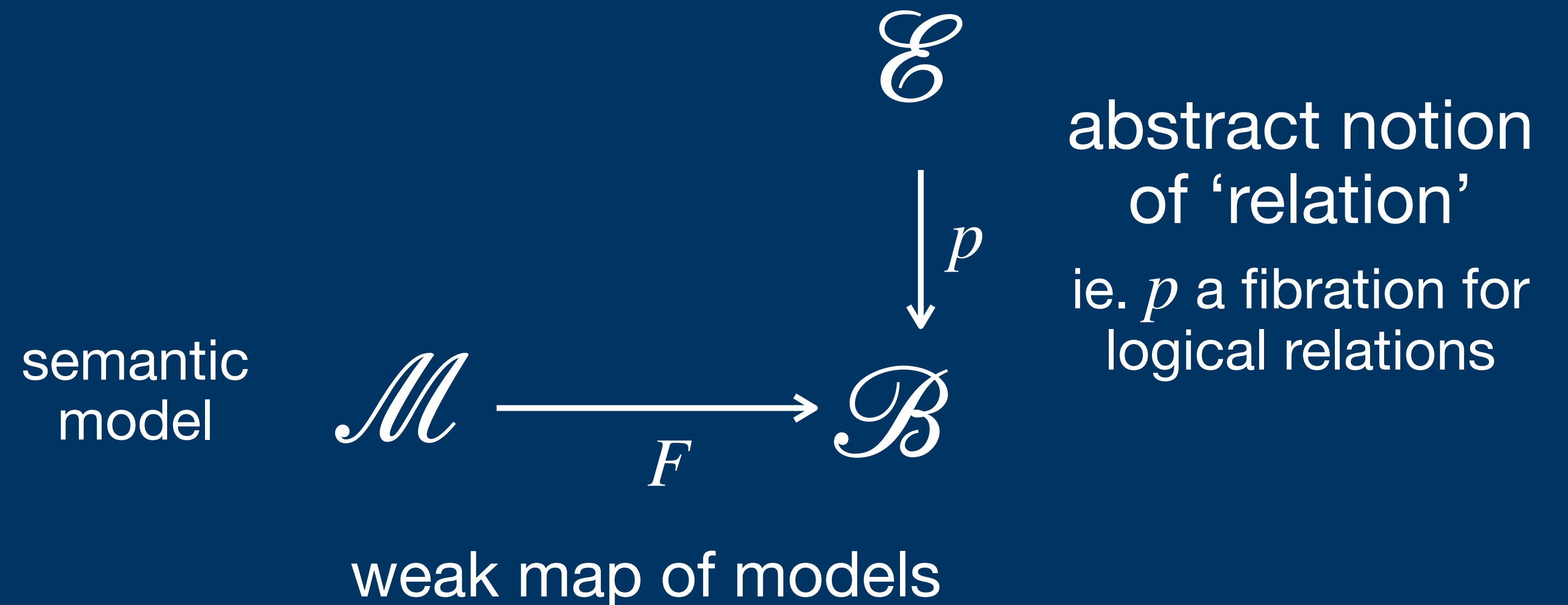
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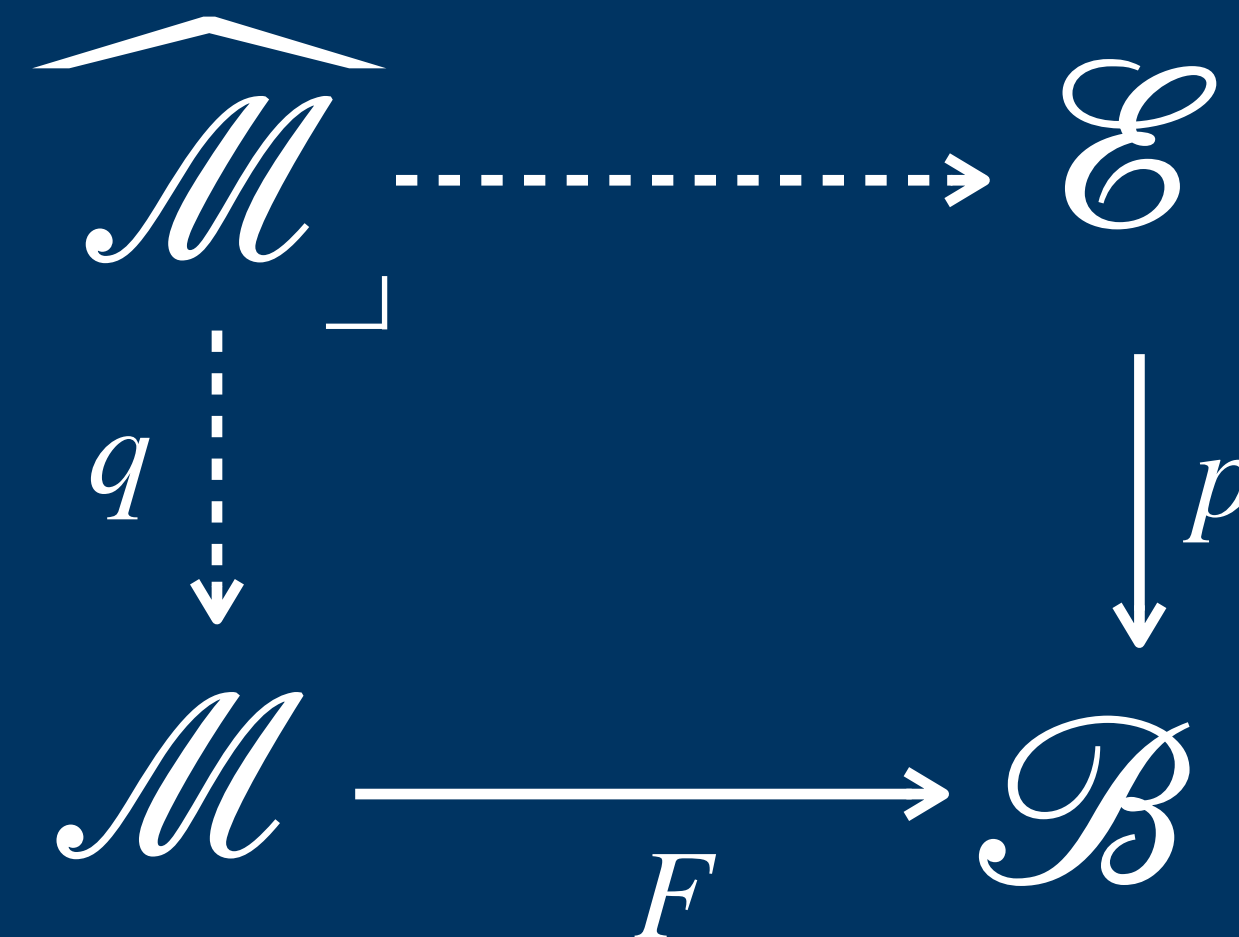
fibration
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lifting theorem:

universal choice of model

‘glueing’ objects of $\mathcal{M}$ to ‘relations’ in $\mathcal{E}$

semantic
model



weak map of models

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objects: $(X \in \mathbf{Set}, R \subseteq X \times X)$

maps: pairs of maps preserving the relation

$$\begin{array}{ccc} \mathbf{BinPred} & \dashrightarrow & \mathbf{Pred} \\ \downarrow q & \lrcorner & \downarrow p \\ \mathbf{Set} \times \mathbf{Set} & \xrightarrow{\times} & \mathbf{Set} \end{array}$$

$$\begin{array}{ccc} \widehat{\mathcal{M}} & \dashrightarrow & \mathcal{E} \\ \downarrow q & \lrcorner & \downarrow p \\ \mathcal{M} & \xrightarrow{F} & \mathcal{B} \end{array}$$

weak map of models

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Logical relations can be seen as...

Level 1: type-indexed families of relations

Level 2: relations models

Level 3: fibrations for logical relations



+ can build new examples
from old ones by glueing

fibration for logical relations:
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Logical relations: a 2-categorical perspective

What's a fibration for logical relations for call-by-push-value?

A **CBPV-model** is a $\mathbf{Psh}(\mathcal{V})$ -enriched adjunction

$$\mathbf{self} \mathcal{V} \begin{array}{c} \xrightarrow{F} \\ \xleftarrow[U]{\perp} \end{array} \mathcal{C}$$

1. What is a **CBPV** fibration for logical relations?
2. Do they satisfy a **lifting theorem**?

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1. What is a **CBPV** fibration for logical relations?
2. Do they satisfy a **lifting theorem**?

can no longer just say a fibration which preserves all the model structure

need a sensible notion of glueing

Observation:

“fibration which preserves model structure”	$\approx$	fibration internal to 2-category of models
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A fibration internal to...

is...

CartCart_{strict}

a fibration strictly preserving products

CCCart_{strict}

a fibration strictly preserving CC-structure

λ_{ml} -models

a fibration strictly preserving λ_{ml} -structure

Suggestion for level 4:

fibration for logical relations	$:=$	fibration internal to 2-category of models
------------------------------------	------	--

*sometimes sufficient but not necessary condition, but a lot of the theory still applies

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Does this work correctly?

Internal fibrations are always fibrations which preserve model structure

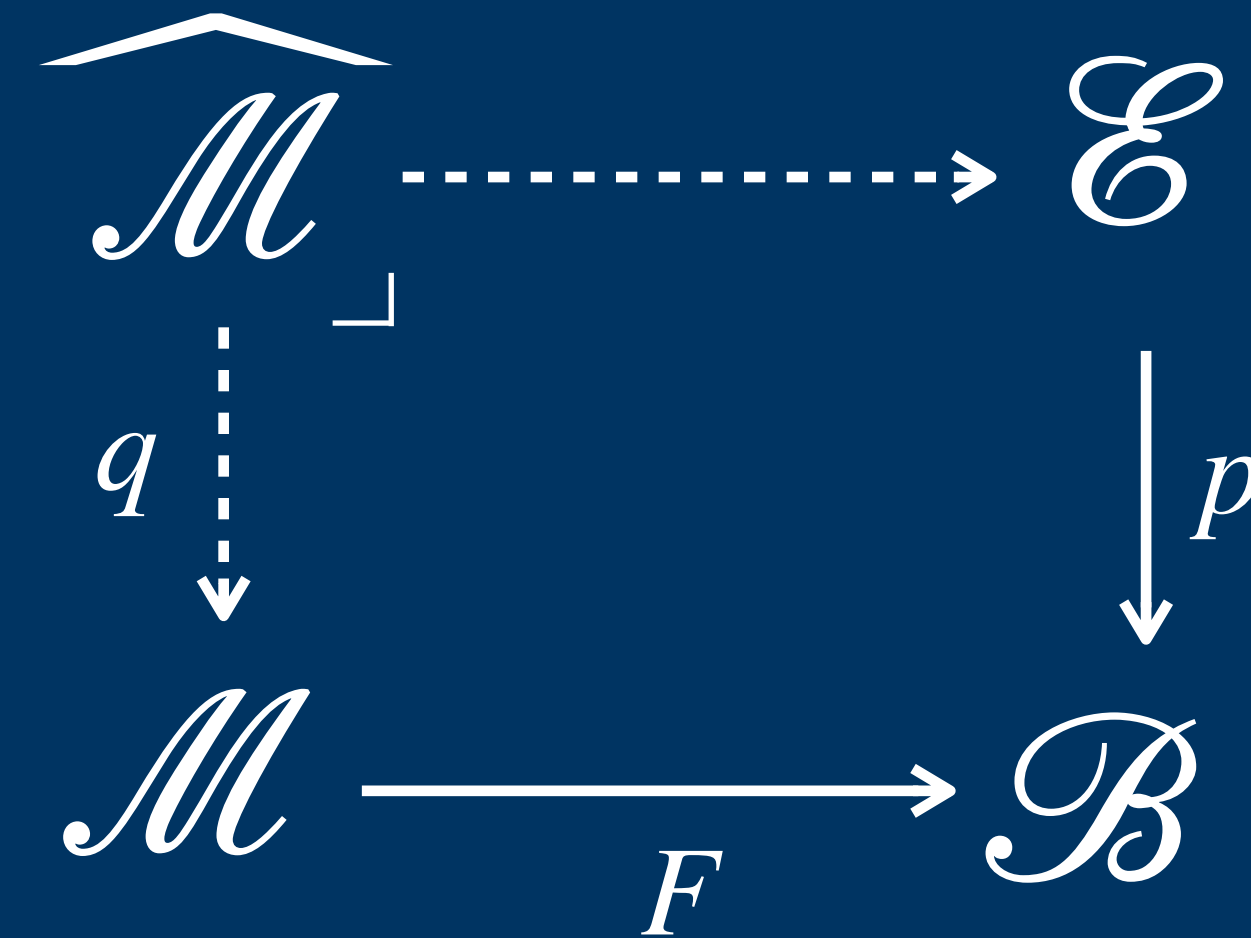
Theorem: let T be a relative pseudomonad along a pseudofunctor $J : \mathcal{A} \rightarrow \mathcal{C}$. Then a fibration in $\text{Ps-}T\text{-Alg}_{\text{ps}}$ is exactly an algebra pseudomorphism $(p, \bar{p})$ such that p is a fibration in $\mathcal{C}$.

Glueing plays well with fibrations

lifting theorem:

universal choice of model
'glueing' objects of $\mathcal{M}$ to 'relations' in $\mathcal{E}$

semantic
model



weak map of models

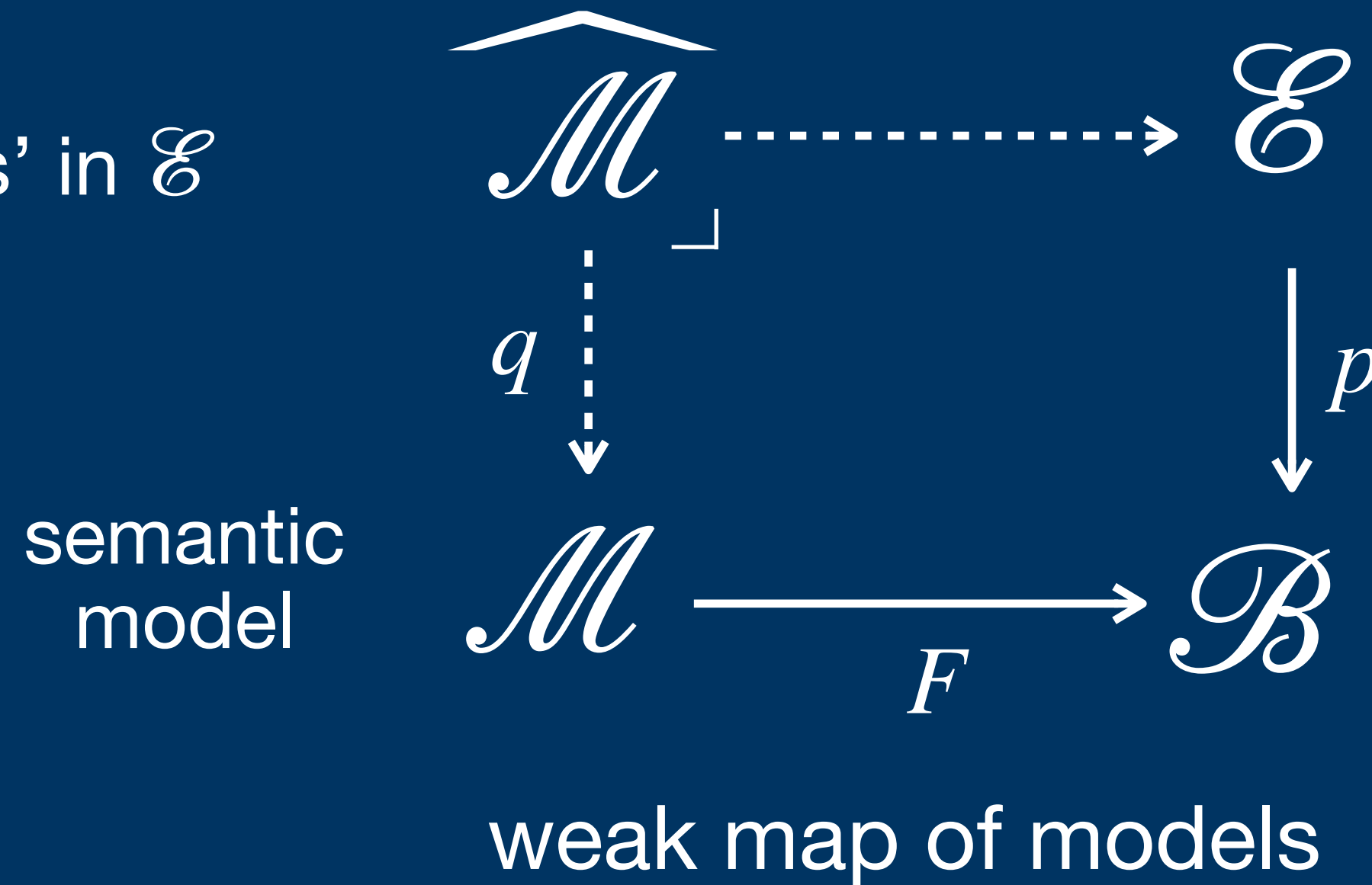
abstract notion
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Proposition [classic]: fibrations are closed under pullback.

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Proposition [classic]: fibrations are closed under pullback.

Glueing reduces to the existence of certain pullbacks in 2-categories of models.

Glueing plays well with fibrations

Proposition: let T be a relative pseudomonad. Consider the following cospan of pseudoalgebras:

$$\begin{array}{ccc} & (E, \dots) & \\ & \downarrow (p, \bar{p}) & \text{lax algebra map +} \\ (A, \dots) & \xrightarrow[(f, \bar{f})]{\text{oplax algebra map}} & (B, \dots) \end{array}$$

fibration

Glueing plays well with fibrations

Proposition: let T be a relative pseudomonad. Consider the following cospan of pseudoalgebras:

$$\begin{array}{ccc}
 (P, \dots) & \dashrightarrow & (E, \dots) \\
 (f^*(p), \text{id}) \downarrow & & \downarrow (p, \bar{p}) \\
 (A, \dots) & \xrightarrow[(f, \bar{f})]{\text{oplax algebra map}} & (B, \dots)
 \end{array}
 \quad \begin{array}{l} \\ \text{lax algebra map +} \\ \text{fibration} \end{array}$$

Then for the pullback $f^*(p) : P \rightarrow A$ in $\mathcal{C}$:

1. P acquires oplax algebra structure,
2. $f^*(p)$ is a strict algebra map, and
3. If $(p, \bar{p})$ is pseudo, this is a 2-pullback in $\text{Oplax-}T\text{-Alg}_{\text{oplax}}$.

Glueing plays well with fibrations
...even when your models are diagrams

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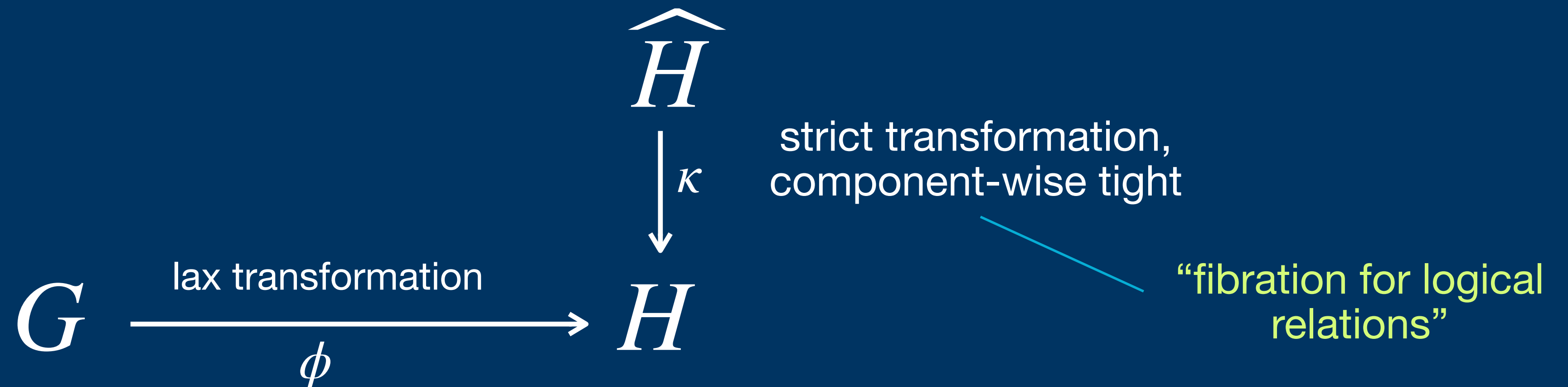
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If **tight** maps in $\mathcal{C}$ are closed under pullback (+ satisfy mild conditions), then in $[\mathcal{I}, \mathcal{C}]_{\text{fax}}$ you can pull a tight map back along any map:

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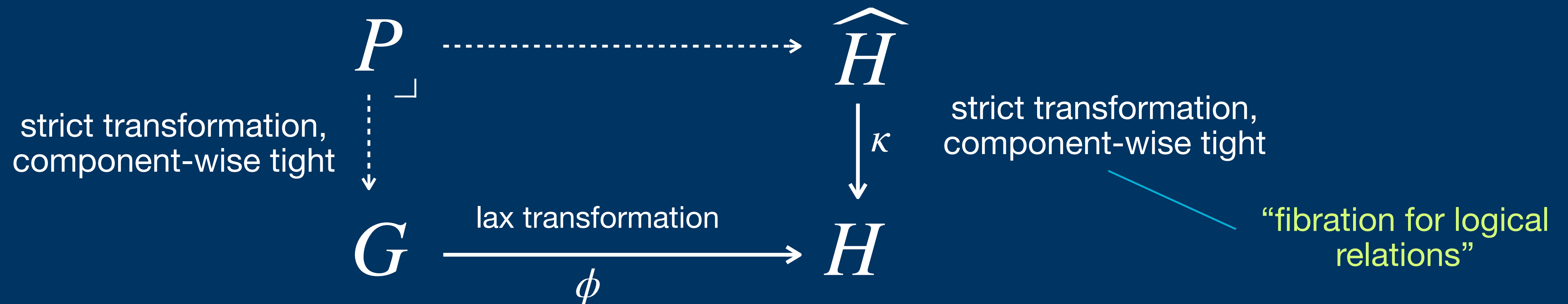
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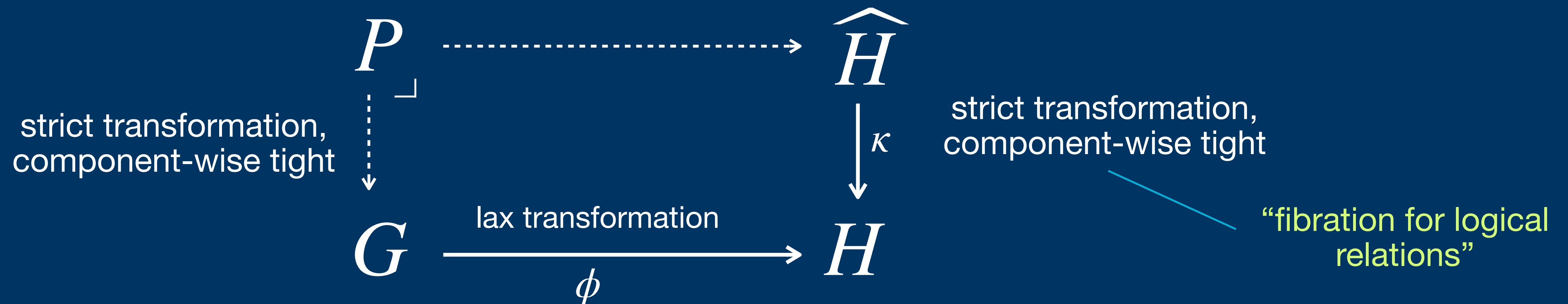
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If **tight** maps in $\mathcal{C}$ are closed under pullback (+ satisfy mild conditions), then in $[\mathcal{I}, \mathcal{C}]_{\text{lax}}$ you can pull a tight map back along any map:



Moral: glueing works for monad models and adjunction models.

under slightly stronger conditions:
can also take fibred products à la
TT-lifting with multiple parameters

1. Fibrations internal to 2-categories tend to be the right thing*.
2. The theory of glueing works as you would hope, even in 2-categories of diagrams.

*sometimes sufficient but not necessary condition, but a lot of the theory still applies

Logical relations for CBPV models

(a recipe)

1. Define a 2-category $\mathbf{LInd}$ of locally-indexed categories,
2. Define CBPV as a sub-2-category of $[\mathbf{Adj}, \mathbf{LInd}]_{\text{ lax }}$,
3. Identify the fibrations internal to $\mathbf{LInd}$,
4. Define a CBPV fibration to be a CBPV-morphism that's component-wise a $\mathbf{LInd}$ -fibration.

logical relations models + a glueing theorem follow from the algebra

a.k.a. Kripke relations of varying arity



Next steps: open logical relations

a.k.a. Kripke relations of varying arity



Next steps: open logical relations

characterise the definable
maps in a model
[cf. Jung & Tiuryn]

semantic NbE à la
*Semantic Analysis of
Normalisation by Evaluation*

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Missing ingredient: a **presheaf** construction for CBPV models

2-categorical statement
of Lafont argument

What about call-by-push-value?

= “locally indexed”

A **CBPV-model** is a $\mathbf{Psh}(\mathcal{V})$ -enriched adjunction

$$\mathbf{self} \mathcal{V} \begin{array}{c} \xrightarrow{F} \\ \xleftarrow[\quad]{\perp} \\ \xleftarrow{U} \end{array} \mathcal{C}$$

What is a ‘presheaf model’ for CBPV?

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First approach: use the enriched presheaf construction.
But: size issues become troubling.

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What is a ‘presheaf model’ for CBPV?

First approach: use the enriched presheaf construction.
But: size issues become troubling.

Alternative: recast CBPV models as certain **fibrations**.

(cf. Ahman, Vákár)

What about call-by-push-value?

each fibre has the same
set of objects, and
reindexing is *ioo*

$\text{Psh}(\mathcal{V})$ -enriched category $\mathcal{C}$ $\overset{\text{folklore(?)}}{\longleftrightarrow}$ 'local' fibration $p : \mathcal{E} \rightarrow \mathcal{V}$

A **CBPV-model** is a (locally) fibred adjunction:

$$\begin{array}{ccc} \text{simple } \mathcal{V} & \begin{array}{c} \xrightarrow{F} \\ \xleftarrow{\perp} \\ \xleftarrow{U} \end{array} & \mathcal{C} \\ & \searrow \text{simple fibration} & \swarrow \\ & \mathcal{V} & \end{array}$$

(cf. Ahman, Vákár)

The 2-category of local fibrations has a Yoneda structure

$$\begin{array}{ccccc}
 \mathcal{E} & & \mathcal{P}(p) & \dashrightarrow & \text{simple}(\text{Psh}(\mathcal{E})) \\
 \downarrow p & \mapsto & \downarrow \lrcorner & & \downarrow \text{simple fibration} \\
 \mathcal{B} & & \text{Psh}(\mathcal{B}) & \xrightarrow{p^*} & \text{Psh}(\mathcal{E})
 \end{array}$$

Induces a presheaf pseudofunctor $\leadsto$ sends adjunctions to adjunctions.

Determines a canonical **presheaf operation on CBPV models**.

Extracting the juice (WIP)

1. Open logical relations for CBPV: characterising definability, semantic NbE.
2. 2-categorical Lafont argument using this construction.
3. Seems to be quite a general way to build presheaf-like pseudomonads.

Summing up

1. Theory of logical relations includes **fibrations for logical relations** and **glueing**.
2. Fibrations **internal to 2-categories of models** seem to capture both aspects quite well.
3. Applies to examples where the models are more complex, e.g. CBPV.
4. Defining an appropriate **presheaf construction** should also lead to open logical relations, definability, NbE, conservativity,

the right
abstraction (?)



follow what
the algebra
tells you



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Can we do the same for other languages?

Suggests a ‘formal theory of logical relations’ + applications.